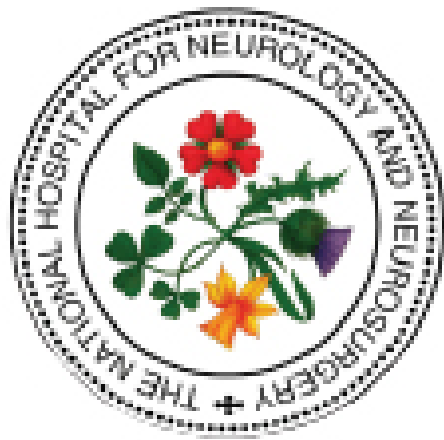




The National Hospital for Neurology and Neurosurgery
Neuroscience Critical Care Unit (NCCU)
Death by Brainstem Death (DBD)

APPROACHING FAMILIES of POTENTIAL DBD DONORS (Collaborative Approach)

If patient meets the following criteria (A + B) please refer to the on-call Specialist Nurse for Organ Donation (SNOD) via the 24-hour pager **0300 20 30 40** to make them aware of a potential donor

A. Absence of one or more cranial nerve reflexes

B. Glasgow Coma Scale (GCS) score of 4 or less that is not explained by sedation

BRAINSTEM DEATH TESTS (BSDT) DISCUSSED with FAMILY
DONATION DISCUSSION IS NOT INITIATED AT THIS TIME

Please ensure that SNOD present at NHNN when brainstem tests due to be carried out

- BSDT performed and results discussed with patient's family
- Doctor to mark patient as deceased on Epic
- Allow time for family to process that death has occurred
- NCC Consultant, NCC nurse, SNOD +/- local faith representative to discuss donation with family
- SNOD gives family information on patient's wishes, where known
- Family supported throughout decision-making process
- Family given relevant information and any questions addressed at this time
- Handprints / hair locks can be offered to family at this time (see SWAN box)
- Death certificate can be completed if **NOT** a coroner's case – NCC medical staff contact Duty Medical Examiner - all NCC deaths discussed with UCLH Medical Examiner
- Medical Examiner: available Monday – Friday contacted via email uclh.medicalexaminer@nhs.net for inpatients at UCLH. For all other patients and during extended hours (Saturdays, Sundays & Bank Holidays 9am-12noon for urgent release due to religious or cultural beliefs and organ donation ONLY) Medical Examiner can be contacted via email uclh.camdenme@nhs.net. Office contact numbers: 020 344(75129) and 020 344(75130) Monday - Friday. During extended hours: 020 3086 7056, 9am-12noon

If family decide donation is not the right option for them, they are thanked for considering donation, and after appropriate time given to family, ventilator and any infusions switched off by senior doctor (ST3 or above)

MONITORING POST DIAGNOSIS OF BRAINSTEM DEATH

1. All neuromonitoring (invasive and non-invasive) should be discontinued – this includes GCS, pupil / limb assessments, RASS, removal of intracranial devices (**This applies to all patients who are diagnosed as brainstem dead**)
2. If the patient is **not** to become an organ donor:
 - All cardiovascular and respiratory observations should cease
 - Monitoring of fluid intake / output should cease
 - EPIC entries should immediately transfer to 'Care following death' documentation

Only exception is if family do not accept validity of BSDT – in this case only neuromonitoring should be discontinued as in 1 above while awaiting a court decision regarding the case

3. If the patient is **to proceed** to organ donation:
 - Continue all cardiovascular and respiratory observations
 - Continue monitoring of fluid intake / output
 - Follow protocol for management of the patient for organ donation with input from SNOD

If family consent to donation the SNOD works with MDT to optimise the patient for donation as per 'Management of Patient For Organ Donation'

SNOD will:

- ▶ Complete Consent and Patient Assessment with family
- ▶ Contact Coroner and GP (where able)
- ▶ Undertake physical examination and assessment of patient
- ▶ Take blood samples for virology and tissue typing
- ▶ Organise CXR, ECG, ECHO (if potential for heart donation)
- ▶ Organise further necessary blood tests and request blood to be cross matched (as required)

The patient is registered with Hub Operations, where matching process will be commenced, and organs offered. The SNOD will liaise with theatres, anaesthetics and organise retrieval teams to attend the hospital

When donation is complete the SNOD performs After Death Care with theatre, organises transfer to the mortuary and arranges for the body to be viewed by the family if desired

SNOD informs the family and staff of outcome of donation and continues post donation support as required

Where consent has been given for tissue donation, retrieval will take place at UCH mortuary (Corneal only donation will take place at NHNN mortuary)

MANAGEMENT of PATIENT for ORGAN DONATION

RESPIRATORY SUPPORT

- Perform lung recruitment manoeuvres
- Review ventilation, ensure lung protective strategy, aim PaO₂ >10kPa with lowest possible FiO₂ (tidal volumes 4-8ml/kg ideal body weight and optimum PEEP 5-10cmH₂O)
- pH >7.25
- Maintain regular chest physiotherapy, including regular suctioning, patient repositioning 2-4 hourly, maintain 35-40 degrees head of bed elevation
- Ensure cuff of endotracheal tube is appropriately inflated
- CT retrieval team will perform bronchoscopy if lungs accept

CARDIOVASCULAR SUPPORT

- Review intravascular fluid status and correct hypovolaemia with fluid boluses
- Commence cardiac output / flow monitoring (FlowTrac)
- Achieve MAP 60-80mmHg (primary goal), CVP 4-10mmHg (secondary goal)
- Commence vasopressin (0.5-4iu/h) where vasopressor required - wean or stop catecholamine vasopressors where able
- Correct coagulopathies / low platelets
- Maintain normothermia (36-37.5 degrees)
- Commence or continue low molecular weight Heparin - ensure use of TEDs / mechanical calf compression

FLUIDS and METABOLIC MANAGEMENT

- Review fluid administration: IV crystalloid maintenance fluid (or NG H₂O where appropriate) to maintain Na⁺ <150mmol/L
- Maintain urine output between 0.5-2.0mL/kg/hr (if >4mL/kg/h consider diabetes insipidus and treat promptly with vasopressor or DDAVP)
- DDAVP 1-4mcg IV titrated
- Insulin to maintain normal blood glucose (4-10mmol/L)
- Continue NG feeding (unless SNOD advises otherwise)

LINES, MONITORING and INVESTIGATIONS (if not already done)

- CVC: right side preferable (Internal jugular or subclavian)
- 12 lead ECG (to exclude Q waves)
- CXR (post-recruitment manoeuvres where possible)
- ECHO if potential for heart donation (where available)
- Review / stop unnecessary medications
- Send Troponin level in all potential heart donors